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SGK1 triggers cartilage degradation in TMJOA via FoxO1/autophagy

Hongyao Tang^{1†}, Chuan Ma^{2†}, Mengchen Xu¹, Gaoyi Wu³, Na Li⁴ and Lei Chen^{1*}

Abstract

Background Temporomandibular joint osteoarthritis (TMJOA) is a prevalent inflammatory disease with unclear pathogenesis. Serum and glucocorticoid-inducible kinase 1 (SGK1) has been implicated in osteoarthritis-related cartilage degeneration. This study investigated the role of SGK1 in a mouse model of TMJOA and explored its mechanism, focusing on SGK1's regulation of Forkhead box O1 (FoxO1) and its subsequent effects on autophagy and extracellular matrix degradation in mouse condylar chondrocytes (MCCs).

Methods A TMJOA model was established using unilateral anterior crossbite (UAC), and an SGK1 knockdown model was generated via intra-articular injection of AAV virus.

Results SGK1 expression was significantly upregulated in condylar cartilage of UAC-induced TMJOA mice. SGK1 knockdown alleviated cartilage matrix degradation and increased the expression of anabolic marker. In MCCs, SGK1 directly bound to FoxO1, promoting its phosphorylation and nuclear export, thereby inhibiting autophagy. Inhibiting SGK1 enhanced autophagy, as evidenced by increased Beclin-1, elevated LC3-II/LC3-I ratio, and decreased P62, which consequently attenuated IL-1 β -induced extracellular matrix catabolism. These effects were reversed by autophagy inhibitor chloroquine and FoxO1 inhibitor AS1842856.

Conclusions This study is the first to propose the regulatory role of SGK1 in TMJOA, providing new insights for future therapeutic research.

Keywords Temporomandibular joint, Condylar cartilage, Chondrocyte, Extracellular matrix, Catabolism, Anabolism

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Background

Temporomandibular joint osteoarthritis (TMJOA) is characterized by the gradual deterioration of cartilage, remodeling of the underlying bone, inflammation in the synovial membrane, and persistent pain [1]. The prevalence of this condition varies from 8% to 60%, with a higher occurrence among women [2, 3]. Currently, symptomatic treatment can control pain and inflammation, but effective strategies for alleviating TMJOA progression are lacking. Unlike the hyaline cartilage found in knees and hips, the TMJ is composed of fibrocartilage and is subjected to a distinct biomechanical environment influenced by complex occlusal forces and mandibular movements [2, 4]. This fundamental anatomical and functional difference suggests that the pathological mechanisms underlying TMJOA may not simply mirror those in other joints, necessitating dedicated investigation.

Autophagy, a crucial cellular catabolic process, has emerged as a key regulatory mechanism in cartilage homeostasis. Its dysregulation is implicated in the pathogenesis of various arthritic degenerative diseases [5, 6]. Under inflammatory conditions, autophagy is often suppressed in chondrocytes, leading to an imbalance in extracellular matrix (ECM) metabolism—promoting the expression of catabolic enzymes (e.g., MMP-13, ADAMTS-5) while inhibiting anabolic factors (e.g., Collagen Type II Alpha 1 Chain, COL2A1; Aggrecan, ACAN) [7–9]. Consequently, enhancing autophagy has been proposed as a potential therapeutic strategy for OA.

Serum and glucocorticoid-regulated kinase 1 (SGK1), a member of the AGC kinase family, is a well-established negative regulator of autophagy [10–12]. Its expression is markedly elevated in osteoarthritic cartilage tissues, wherein SGK1 suppression effectively activates autophagy and significantly attenuates ECM degradation [13]. In chondrocytes, inhibition of SGK1 with the Chinese herbal compound columbianetin activates autophagy, thereby reducing cell death and alleviating osteoarthritis progression [14]. Moreover, SGK1 modulates autophagy through its action on the downstream transcription factor Forkhead box O1 (FoxO1) [15–17]. SGK1 phosphorylates FoxO1, prompting its translocation from the nucleus to the cytoplasm, which leads to FoxO1 down-regulation and consequent suppression of autophagy [18, 19].

Given the potential involvement of SGK1-FoxO1 signaling and autophagic dysfunction in TMJOA, we hypothesized that SGK1 exacerbates TMJOA by inhibiting autophagy via FoxO1 phosphorylation. In this study, we aimed to understand the impact of SGK1 on TMJOA in a mouse model. To elucidate potential underlying mechanisms, we investigated how SGK1 affects autophagy and cell matrix degradation in mouse condylar chondrocytes (MCCs) by regulating FoxO1. Our findings

contribute toward developing more effective TMJOA treatments.

Methods

Animal models

Twelve C57BL/6J mice (male; 8 weeks) were divided into Control ($n=6$) and unilateral anterior crossbite (UAC) groups ($n=6$) using a random allocation method. In the UAC group, the upper and lower incisors on the right side were bonded to construct an anterior anticlinal occlusal relationship, and the intervention lasted for 21 days. In the Control group, the same operation was performed but without bonding the cannulas. All mice were reared under the same conditions. All experimental procedures involving animals were approved by the Ethics Committee at Shandong University (no: 2022080606, 29 July 2022).

Histological analysis

Arthritis was assessed using the Osteoarthritis Research Society International (OARSI) scoring system: grade 0: normal cartilage with smooth surface; grade 1: cartilage surface affected, with slight fibrosis and abrasion-like changes; grade 2: localized fibrosis in superficial layers of cartilage; grade 3: presence of vertical fissures (3.5 if there are branches); grade 4: cartilage erosion (4.5 if there is exfoliation); grade 5: exposed or semi-exposed phenomenon in the depths of cartilage; repair tissue on bone surface may be 5.5; grade 6: joint deformation with marginal or central osteophytes.

To analyze optical density, positive cellular responses (brownish-yellow or reddish area) in the image were assessed in Image-Pro Plus 6.0 (Media Cybernetics, Inc. Rockville, MD, USA). The total positive optical density was determined by multiplying the area of the image pixels, and the average optical density (AOD) was obtained through their division. Positive response increases as the AOD rises, which is calculated by dividing optical density by pixel area.

Hematoxylin-eosin (H&E) staining

Tissue sections were placed in xylene for dewaxing; hematoxylin staining was processed for 5 min. After acid washing, the sections were immersed in a 0.5% eosin solution for 2 min before being desiccated and securely sealed. Sections were analyzed using the OARSI scoring system.

Safranin O staining

The tissue sections underwent deparaffinization, followed by 3–5 min of staining with solid green and subsequent washing to eliminate excess stain. The sections were stained with safranin O for 1 min and washed again, dehydrated in anhydrous ethanol, and blocked with

neutral gum. The proportion of positive areas was analyzed based on AOD.

Immunohistochemical (IHC) staining

Sections were dewaxed and dehydrated and then immersed in a 3% hydrogen peroxide solution for approximately 10 min, followed by rinsing with PBS. An antigen retrieval solution was added and incubated at ambient temperature for 20 min; 10% goat serum was used for sealing. The primary antibody was diluted and incubated at 37 °C for 2 h. The secondary antibody labeled with horseradish peroxidase (Beyotime, China) was added and incubated at 37 °C for 1.5 h, and chromatography was performed. The proportion of positively stained cells was analyzed based on AOD, and the primary antibodies were against SGK1 (1:1000, Sigma, USA) and COL2A1 (1:1000, Sigma).

SGK1-knockout mice

Twenty-four male, 6–8-week-old C57BL/6 mice were selected for TMJ intra-articular injection of AAV9-SGK1 with AAV9-GFP (Vizen Biotech, China). Each mouse was injected unilaterally into the TMJ with 15 μ l, 1×10^{12} vg/ml. After 2 weeks, a 3-week UAC inflammation model (UAC group) and negative control groups were constructed. The mice were categorized into AAV9-GFP (6 mice), AAV9-SGK1 (6 mice), UAC + AAV9-GFP (6 mice), and UAC + AAV9-SGK1 (6 mice) groups. The final TMJs were collected for histological analysis after euthanizing the mice.

MCCs culturing and characterization

MCCs were isolated from condylar tissues of eight male, 3-week-old C57BL/6J mice. The condylar tissues were fragmented using delicate surgical scissors; the condylar tissues underwent enzymatic digestion with 1 ml of pancreatic enzyme solution (0.25%) at 37 °C for 20 min, followed by digestion with 1 ml of type II collagenase solution (2 mg/ml) at 37 °C for 120 min. Afterward, the cells were placed in a culture medium containing 15% fetal bovine serum and incubated at 37 °C with 5% CO₂. To visualize their characteristics, the MCCs were stained with Alcian blue (1%) for approximately 30 min, after which they were rinsed with distilled water. Samples were observed following dehydration and transparency procedures.

MCCs were subsequently subjected to procedures and assays, including cell transfection, drug treatment, cell immunofluorescence, monodansylcadaverine (MDC) staining, real-time quantitative PCR, western blotting, and immunoprecipitation, which are detailed in Additional File (Table S1, Table S2 and Supplementary Methods).

Cell transfection

After achieving 85% density in 6-well plates, the MCCs underwent transfection using a transfection reagent (lipofectamine RNAiMAX reagent) (GenePharma, China). The cells were collected 48 h after transfection for examination. si-RNA (GenePharma) sequences are shown in Table S1.

Autophagy and FoxO1 Inhibition

To investigate functional rescue, we administered specific inhibitors to MCCs following siRNA transfection and IL-1 β challenge. The cells were incubated with either chloroquine (CQ, 25 μ mol/L; Sigma) to suppress autophagy or AS1842856 (1 μ mol/L; Sigma) to inhibit FoxO1 signaling for a duration of 24 h. Additional experimental procedures are described in the Supplementary Materials.

Statistical analysis

The t-test was used to analyze the differences between two groups. Data across multiple groups were compared using a one-way ANOVA followed by the Student–Newman–Keuls post-hoc test. The normality was confirmed using Shapiro–Wilk test, and homogeneity of variance was verified via F-tests (for t-tests) or Bartlett's test (for ANOVA). * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$ indicated significance. All statistical tests were performed using GraphPad Prism 9.0 (La Jolla, CA, USA).

Results

Establishment of TMJOA model and upregulation of SGK1

H&E staining and OARSI scoring indicated that the UAC group had a significantly higher score than the Control group (Fig. 1A and B). Safranin O staining analysis of AOD demonstrated reduced proteoglycan content in the UAC group (Fig. 1C and D). IHC staining and optical density values showed that the expression rate of COL2A1 in the UAC group was significantly lower than that in the Control group (Fig. 1E and F). Thus, the UAC inflammation model was successfully constructed, and the mouse condyle cartilage matrix degraded under inflammatory conditions. Conversely, IHC indicated markedly higher SGK1 expression in the UAC group (Fig. 1G and H). MCCs were successfully isolated and identified via Alcian blue staining (Fig. 1I). Furthermore, in IL-1 β -stimulated MCCs, IF, qPCR, and WB analyses consistently showed upregulation of SGK1 at both mRNA and protein levels (Fig. 1J–N). These results confirm aberrant SGK1 upregulation in TMJOA inflammation.

SGK1 knockdown alleviates TMJOA cartilage ECM degradation

To explore the role of SGK1 in TMJOA, SGK1-knockdown mice were generated via articular cavity injection

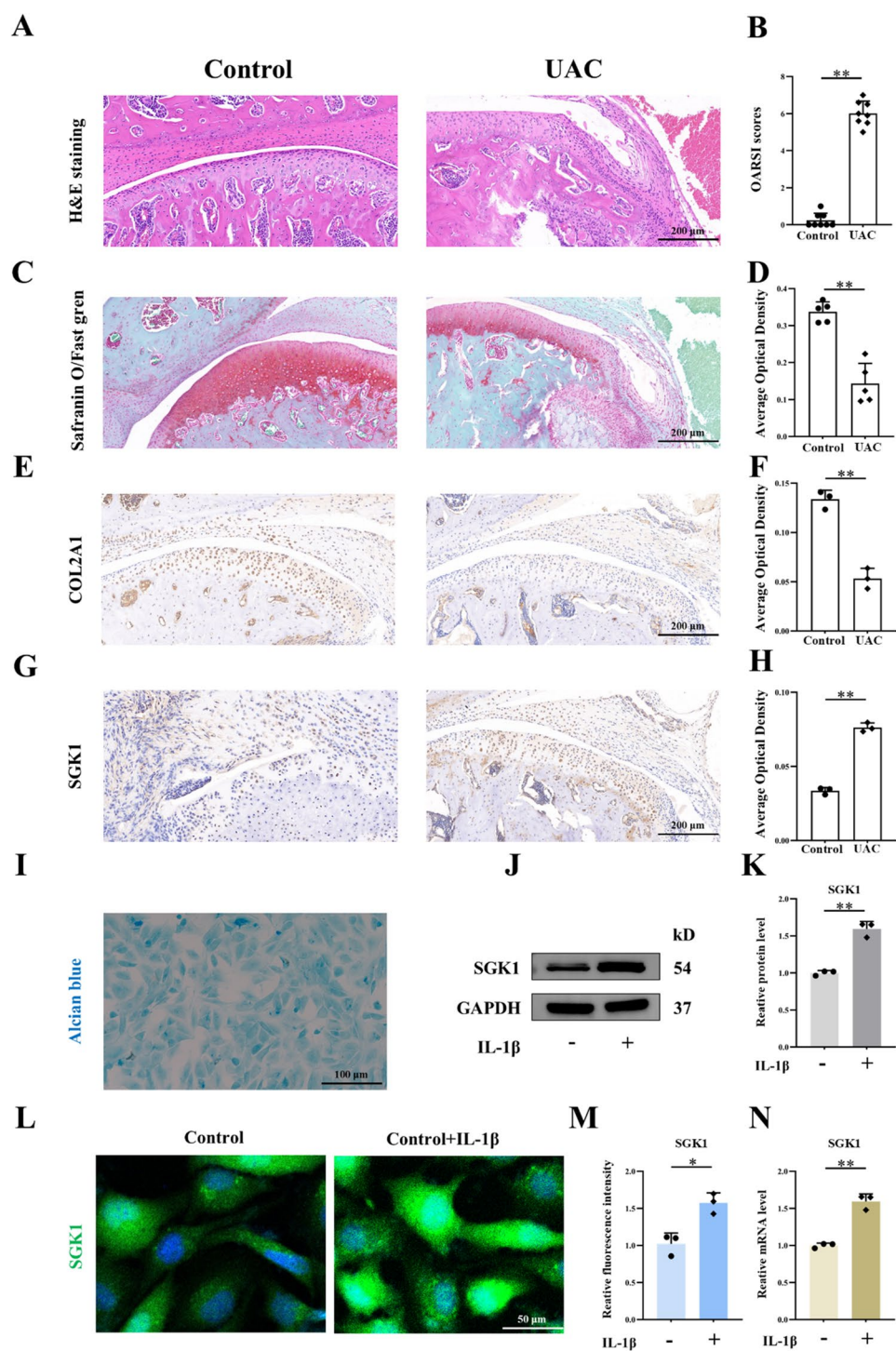


Fig. 1 SGK1 expression in TMJOA cartilage and MCCs. **A, C** Representative H&E and Safranin O staining of normal and UAC inflamed TMJ (scale bar: 200 μ m). **B** Cartilage destruction in Control and UAC groups, under H&E staining, assessed based on OARSI scoring. **D** Cartilage matrix degradation in Control and UAC groups, under saffron O staining, assessed based on optical density value. **E** Representative IHC images of COL2A1 (**E**) and SGK1 (**G**) in the above TMJ (scale bar: 200 μ m). Optical density quantification of COL2A1 (**F**) and SGK1 (**H**) (Control vs. UAC groups, IHC). **I** Representative Alcian blue staining of extracted MCCs (scale bar: 100 μ m). **J, K** WB of MCCs after 24 h IL-1 β treatment. **L, M** IF results after 24 h of IL-1 β -treated MCCs. **N** qPCR of MCCs after 24 h IL-1 β treatment. Data represent means $\pm$ SD ($n \geq 3$), * $P < 0.05$, ** $P < 0.01$

of AAV9 and subjected to UAC-induced inflammation (Fig. 2A). SGK1 knockdown efficiency was verified via IHC (Fig. 2B and C). Safranin O staining revealed greater cartilage matrix loss in the UAC + AAV9-GFP group than in the UAC + AAV9-SGK1 group (Fig. 2D–F), suggesting that SGK1 knockdown alleviates matrix degradation. Under inflammatory conditions, SGK1 knockdown significantly increased COL2A1 expression (Fig. 2F and G), whereas no notable difference was observed between non-inflammatory AAV9-GFP and AAV9-SGK1 groups, indicating that SGK1 mainly influences ECM metabolism under inflammatory stimulation.

Given that SGK1 knockdown could alleviate ECM degradation in TMJOA, we knocked down SGK1 in MCCs with the IL-1 β -induced inflammatory microenvironment (Fig. 2H–J). qPCR and WB showed that, in this inflammatory context, MCCs downregulated COL2A1 and ACAN and upregulated MMP13 and ADAMTS-5, i.e., exhibiting ECM degradation. Knocking down SGK1 gradually restored ECM metabolic homeostasis (Fig. 2K–M). Thus, at the cellular level, IL-1 β -induced ECM degradation occurred in the inflammatory microenvironment of MCCs, which was reversed by SGK1 knockdown.

SGK1 knockdown enhances autophagy to alleviate ECM degradation

As SGK1 acts as an autophagy inhibitor, we investigated whether cellular autophagy contributes to SGK1-mediated ECM regulation. SGK1 knockdown promoted autophagy in MCCs, which was confirmed by the relative upregulation of Beclin-1 and LC3-II/LC3-I conversion levels and the relative downregulation of P62 expression (Fig. 3A–C). MDC staining further confirmed increased autophagosome formation (Fig. 3D and E), indicating that SGK1 negatively regulates autophagy in MCCs. Subsequently, an autophagy inhibition rescue experiment was conducted using CQ, with ECM metabolic indicators detected using WB. CQ reversed the relief of ECM metabolic disorders originally caused by SGK1 knockdown (Fig. 3F and G). This functional reversal experiment further demonstrated that the ECM metabolic imbalance exhibited in inflamed MCCs is mediated by autophagy, and that SGK1 also affects ECM metabolism through autophagy.

SGK1 phosphorylate FoxO1 and regulate its subcellular localization

Co-immunoprecipitation confirmed a direct interaction between SGK1 and FoxO1 in MCCs (Fig. 4A). WB confirmed that SGK1 knockdown inhibited FoxO1 phosphorylation in the inflammatory microenvironment of MCCs, i.e., the ratio of p-FoxO1/FoxO1 was markedly downregulated (Fig. 4B and C). Phosphorylation regulation is often associated with subcellular localization.

Our WB with IF analyses showed that SGK1 knockdown increased FoxO1 in the nucleus (Fig. 4D–G). In the inflammatory microenvironment of MCCs, SGK1 interacts with FoxO1 and induces phosphorylation, thereby regulating the subcellular localization of FoxO1. Specifically, this leads to an increase in FoxO1 outside the nucleus.

To determine whether FoxO1 mediates SGK1-regulated autophagy and ECM metabolism, rescue experiments were performed using a FoxO1 inhibitor (AS1842856). WB showed that SGK1 knockdown alleviated ECM degradation and promoted autophagy in MCCs, which was reversed by AS (Fig. 5A and B). Thus, in the inflammatory conditions, SGK1 knockdown increased cellular autophagy, and alleviated ECM degradation in a FoxO1-dependent manner.

Discussion

Current research investigating the influence of SGK1 on OA progression is focused on major joints like knees and ankles and less so on TMJOA. As the only joint in the maxillofacial region, the soft and hard tissues of the TMJ are constantly altered by the interactions with the surrounding structures and their environment, including occlusal forces, resulting in a landscape more complex than that found in other joints. Additionally, the sensitivity of SGK1-related inhibitors or drugs is regulated to some extent by SGK2 and SGK3 [1, 20]. In drug trials, the treatment of SGK1 by relevant inhibitors or drugs had a limited impact on the normal homeostasis of the overall organism [21]. Therefore, exploring the link between SGK1 and TMJOA has potential clinical therapeutic value.

In this study, SGK1 was found to regulate autophagy during cartilage matrix metabolism by altering FoxO1 subcellular localization. We verified that SGK1 knockdown can alleviate ECM degradation in TMJOA by activating cellular autophagy through both in vivo and ex vivo experiments. Rescue experiments using a FoxO1-specific inhibitor (AS1842856) confirmed that FoxO1 is the core downstream target of SGK1 in regulating cartilage matrix metabolism (Fig. 5A and B). Furthermore, its mediated decrease in FoxO1 phosphorylation, resulting in increased retention in the nucleus, may be the key to the regulation of ECM metabolism by autophagy-associated proteins in MCCs (Fig. 5C).

SGK1 is an immediate-early gene widely expressed in the tissues examined but is subject to strict transcriptional and post-transcriptional regulation. The half-life of both the mRNA and protein does not exceed 30 min [22]. This early and stringent transcriptional management means that SGK1 expression is characterized by protein stability, and therefore, any expression aberrations may influence various pathophysiological processes. SGK1

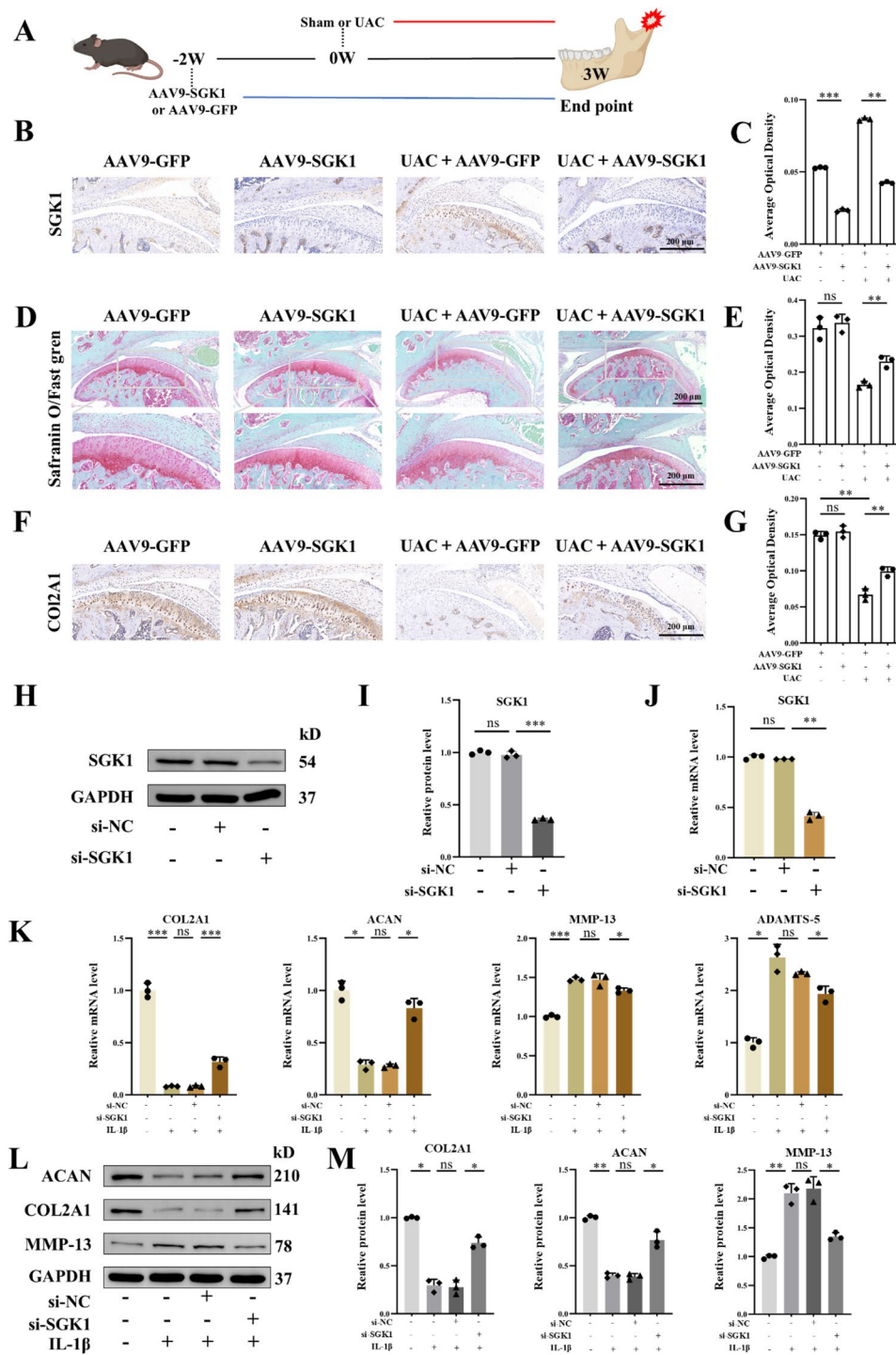


Fig. 2 Effect of SGK1 knockdown on TMJ cartilage matrix metabolism. **A** Schematic diagram of animal experiments. **B** Representative IHC images of SGK1 in the four groups. **C** Quantitative analysis of AOD. **D** Representative images of Sanguinaria O staining in the four groups and **(E)** quantitative analysis of AOD, scale bar: 200 μ m. **F** Representative images of cartilage matrix metabolism index (COL2A1) IHC and **(G)** quantitative analysis of AOD, scale bar: 200 μ m. **H–J** Validation of SGK1 knockdown efficiency in transfected MCCs. **K** qPCR for COL2A1, ACAN, MMP13, ADAMTS-5 in MCCs (transfected 24 h, then IL-1 β treated 24 h). **L** WB for COL2A1, ACAN, MMP13 under the same conditions as **(K)**. **M** Semi-quantitative analysis represented as the average $\pm$ standard deviation ($n = 3$). * $P < 0.05$, ** $P < 0.01$, ns indicates no significant difference

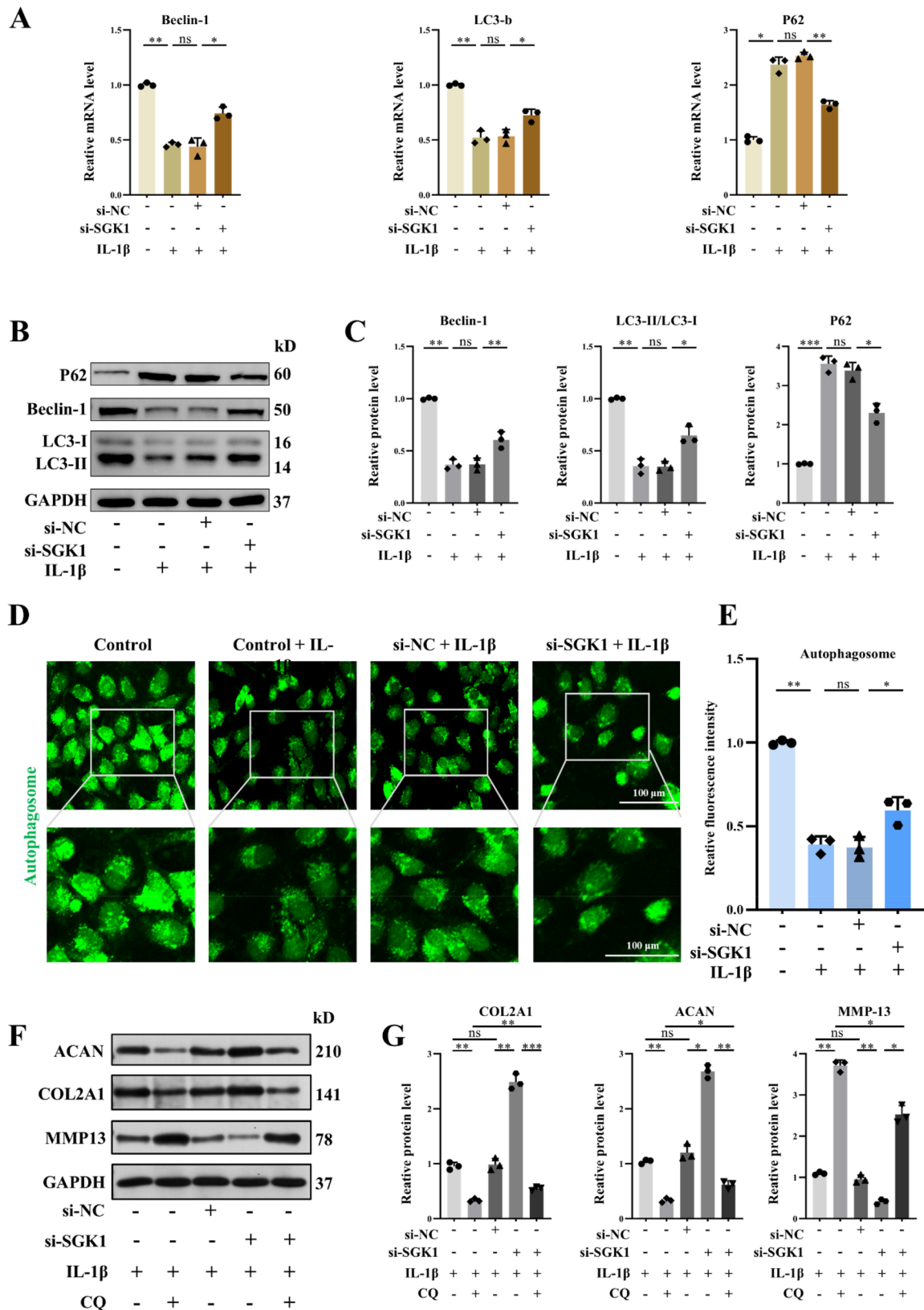


Fig. 3 SGK1-mediated autophagy in MCCs affects ECM metabolism. **A** qPCR for autophagy markers in MCCs (transfected 24 h, then IL-1 β treated 24 h). **B** WB for autophagy markers under the same conditions as **(A)**. **C** Semi-quantitative analysis of **(B)**. **D**, **E** MDC staining to detect autophagic vesicles in MCCs. **F** WB for ECM metabolism markers in MCCs after 24 h CQ treatment. **G** Semi-quantitative analysis of **(F)**. Data represent means $\pm$ SD ($n=3$), $*P<0.05$, $**P<0.01$, $***P<0.001$, ns no significant difference

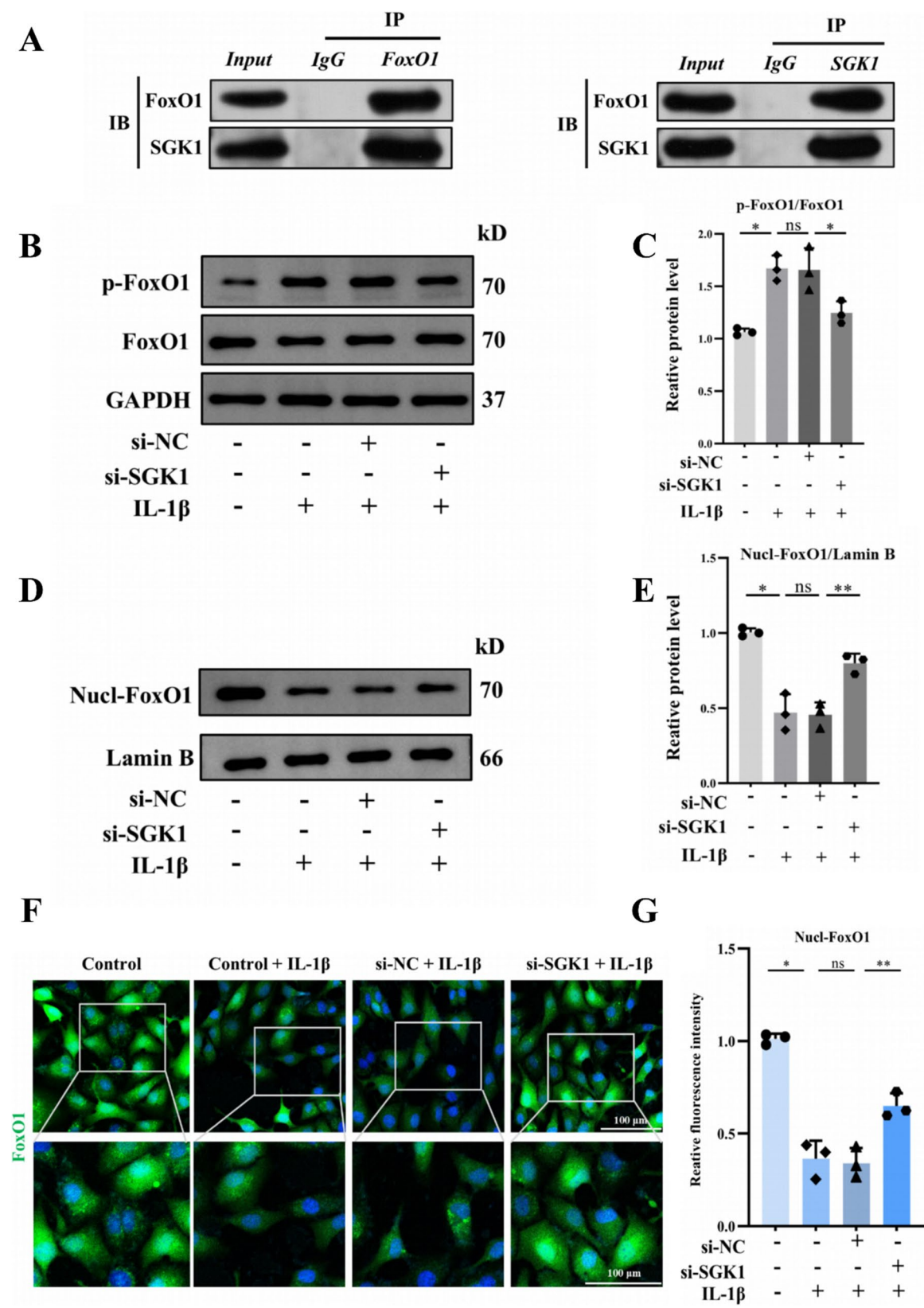


Fig. 4 SGK1 binds FoxO1 to phosphorylate it and regulate its subcellular localization. **A** Co-IP verified the interaction between FoxO1 and SGK1. **B** After 24 h of treatment with si-NC or si-SGK1 transfection, followed by 2 h of treatment with IL-1 β , the expression of FoxO1 and p-FoxO1 was detected via WB. **C** Semi-quantitative analysis. **D** FoxO1 expression levels in the nuclei of MCCs detected via WB. **E** Semi-quantitative analysis. **F, G** Expression of FoxO1 in the nucleus of cells detected via IF. Data represent means $\pm$ SD ($n=3$), * $P<0.05$, ** $P<0.01$, ns no significant difference

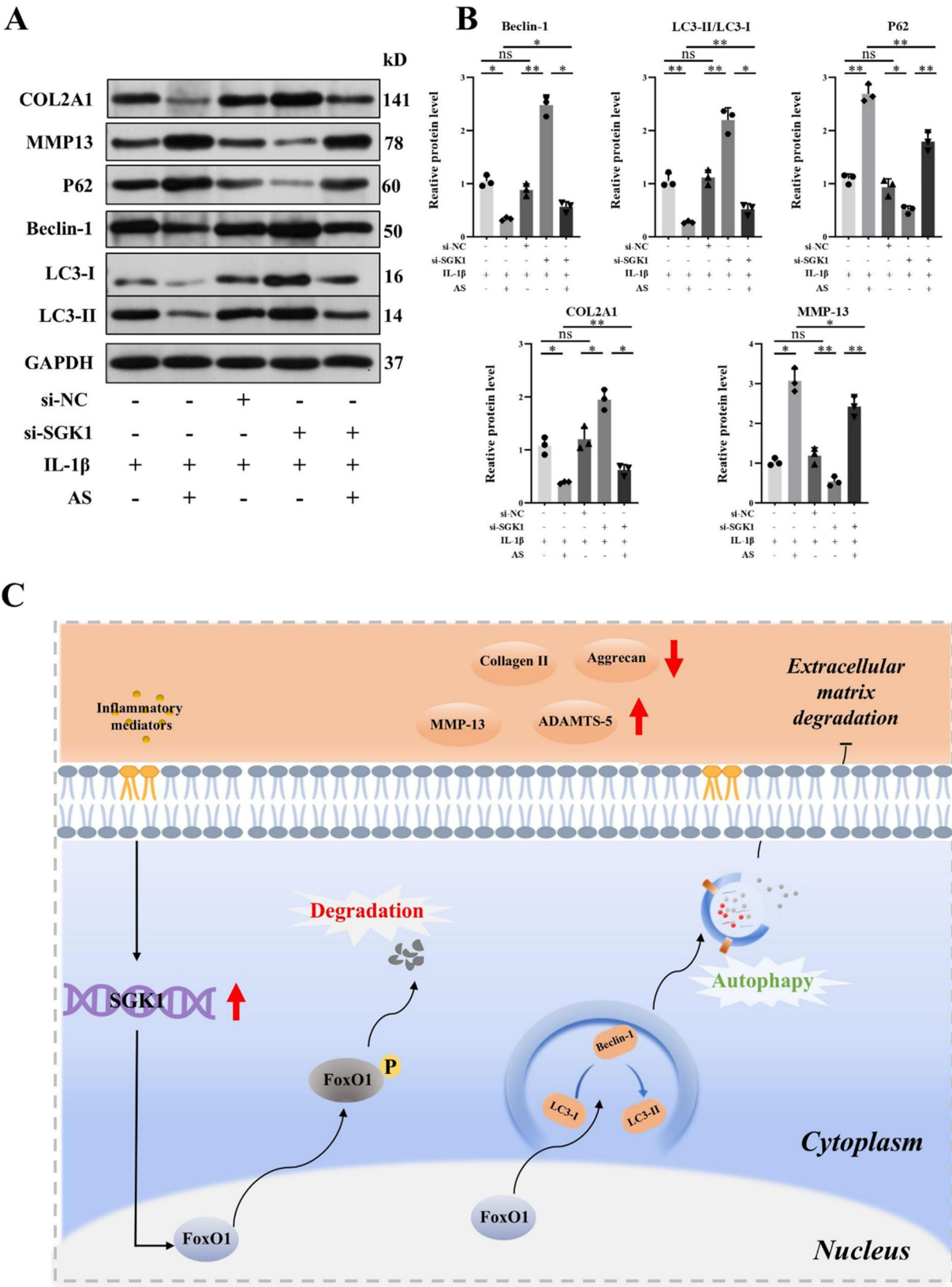


Fig. 5 SGK1 regulates autophagy and alleviates ECM degradation in MCCs via FoxO1. **A** Autophagy and ECM metabolism indexes were detected via WB after transfection for 24 h, underwent continued treatment with IL-1 β for 24 h, and were finally treated with AS for 24 h. **B** Semi-quantitative analysis. Data represent means $\pm$ SD ($n=3$), * $P<0.05$, ** $P<0.01$, ns no significant difference. **C** Mode of action diagram of SGK1 regulating autophagy level through FoxO1 to alleviate cartilage extracellular matrix degradation in TMJOA, “ $\rightarrow$ ” indicates “promote”, and “ $\perp$ ” indicates “inhibit”

expression is generally minimal in most normal cells; however, it undergoes a significant increase under specific pathophysiological conditions such as excessive administration of glucocorticoids or saline corticosteroids, inflammation accompanied by the release of TGF- β , elevated blood sugar levels, cellular shrinkage, and ischemia [23]. SGK1 overexpression and aberrant activity are closely associated with various disease mechanisms, including immunoinflammation, hypertension, and fibrosis [21]. Aberrant upregulation of SGK1 has been observed in various chronic inflammatory diseases, including chronic inflammatory nephropathy [24] and pulmonary fibrosis [25]. Therefore, in our preliminary study, SGK1 upregulation in TMJOA was confirmed both in vitro and in vivo.

In inflammatory diseases, pathologic alterations in the ECM microenvironment are common. In normal TMJ, the cartilage ECM is structurally stable [26]. The development of TMJOA is often accompanied by dynamic imbalance, degradation, and disruption of the cartilage ECM structure [27]. Therefore, we linked SGK1 to cartilage ECM metabolism and confirmed that SGK1 knockdown attenuated ECM metabolic imbalance. This implies that SGK1 is involved in TMJOA development.

SGK1 is a regulator of cellular autophagy [28, 29], and suppressing SGK1 using SI113 enhances the conversion of LC3-II to LC3-I and elevates Beclin-1 levels in human glioblastoma cells, thereby facilitating cellular autophagy [30, 31]. However, its role in the regulation of autophagy proteins in MCCs is unclear. We confirmed that the cellular autophagy mechanism mediated by SGK1 is also present in MCCs and that the ECM degradation exhibited during MCC inflammation could be regulated by autophagy mechanisms. Therefore, the regulatory mechanism of SGK1 in MCC inflammation requires further study.

FoxO1 and SGK1 interact to regulate autophagy in OS cell lines [32, 33], but this has not been confirmed in MCCs. In this study, we determined that p-FoxO1 undergoes direct binding with SGK1 in MCCs and subsequently influences its subcellular localization. FoxO1 is rendered inactive through phosphorylation, causing its translocation from the nucleus to the cytoplasm and subsequent degradation. Thus, it loses its ability to regulate autophagy-related genes like Beclin-1 and LC3-b. This could potentially explain how SGK1 influences autophagy in MCCs. The potential alteration in the intracellular distribution of FoxO1 may be associated with the presentation of its phosphorylated modification to chaperone 14-3-3 at its binding site, leading to their simultaneous movement into the nucleus. The exposure of the 14-3-3 chaperone protein modifies the nuclear localization sequence of FoxO1 and impedes its re-entry into the nucleus [34].

This study presents findings on the correlations among SGK1, autophagy, and cartilage ECM metabolism in TMJOA pathology. SGK1 is upregulated during TMJOA, and SGK1 knockdown mediates autophagy via the subcellular localization of FoxO1, which alleviates ECM degradation. The properties of SGK1 inhibitors observed in their respective sensitization assays suggest the potential for “SGK1 inhibition” to be investigated as a viable clinical approach for treating TMJOA. However, the lack of functional response validation in animal studies and the lack of sufficient human TMJ samples have limited further exploration of the association between SGK1 and TMJOA in the present study.

Conclusions

This study reveals for the first time the mechanism by which SGK1 acts as a regulatory factor involved in the TMJOA process in MCCs. SGK1 can directly bind to FoxO1 and enhance the p-FoxO1/FoxO1 translocation as well as nuclear degradation, thereby inhibiting autophagy in MCCs and disrupting the homeostasis of the cartilage matrix. Therefore, it may be possible to halt TMJOA progression via SGK1 inhibition.

Abbreviations

ACAN	Aggrecan
AOD	Average optical density
COL2A1	Collagen type II alpha 1 chain
ECM	Extracellular matrix
FoxO1	Forkhead box O1
H&E	Hematoxylin-eosin
IHC	Immunohistochemical
MDC	Monodansylcadaverine
OARSI	Osteoarthritis research society international
SGK1	Serum and glucocorticoid-inducible kinase 1
TMJOA	Temporomandibular joint osteoarthritis
UAC	Unilateral anterior crossbite

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13075-025-03690-w>.

Additional file 1

Additional file 2

Additional file 3: Table S1 Sequences of si-RNA primers. Table S2 RT-qPCR primer sequences. Supplementary Methods.

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Authors' contributions

L. Chen contributed to conception, design, data acquisition, and analysis and drafted and critically revised the manuscript; H. Tang contributed to design and data interpretation and drafted the manuscript; C. Ma contributed to data interpretation and critically revised the manuscript; M. Xu contributed to acquisition and interpretation, and drafted the manuscript; G. Wu, contributed to data interpretation, and critically revised the manuscript; N. Li, contributed

to data interpretation, and critically revised the manuscript. All authors gave their final approval and agree to be accountable for all aspects of the work.

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Data availability

All data generated or analysed during this study are included in this published article [and its supplementary information files].

Declarations

Ethics approval and consent to participate

All experimental procedures involving animals were approved by the Ethics Committee at Shandong University (no: 2022080606, 29 July 2022). The research methods in this article conform to the ARRIVE Guidelines and the completed ARRIVE 2.0 checklist (full version) uploaded as Supplementary Information.

Consent for publication

Not Applicable.

Competing interests

The authors declare no competing interests.

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